

Cultural Heritage Exploration through LLM-Enabled Question Answering and Quiz Interaction in Mobile Augmented Reality

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ABSTRACT

This study presents an LLM-based augmented reality (AR) system designed to enhance cultural heritage exploration through natural language question answering (QA) and quiz-based interactions. A structured knowledge base guides the LLM to provide context-aware explanations in response to user queries, while generated quiz items transform cultural heritage data into interactive learning content within an AR environment. The system supports in-situ and map-based exploration, enabling users to review and learn from questions generated by others, fostering collective knowledge building. This design demonstrates the potential of combining LLM-driven interaction with AR to enrich interpretation, support learning, and promote cultural heritage experiences.

Index terms: Mobile AR (MAR), Large Language Model (LLM), Retrieval-Augmented Generation (RAG), Cultural Heritage, Context-Aware Learning.

1 INTRODUCTION

Augmented Reality (AR) enhances the real world by overlaying digital content and has become a valuable tool in cultural heritage for delivering reconstructed artifacts, spatially aligned information, and contextual explanations. Mobile AR (MAR) using smartphones has further increased accessibility and supported immersive experiences through sensory engagement, storytelling, behavioural guidance, and gamification [1]. Recent work has shown growing interest in integrating large language models (LLMs) into XR environments, particularly for educational settings where LLMs offer adaptive feedback and personalized guidance, thereby amplifying the interpretive and learning potential of MAR-based cultural heritage applications [2].

In educational settings, LLMs have proven valuable by analyzing learner input in real-time and delivering personalized feedback [2]. This capability significantly enhances the learning experience in various domains, including language acquisition, simulation-based training, XR-supported learning, and virtual classroom environments. In the cultural heritage domain, XR-LLM research has mainly focused on personalized interpretation, narrative explanation, and recommendation [3, 4].

However, existing MAR systems still rely on static information delivery, offering limited support for interactive or context-sensitive learning. Likewise, most current LLM in XR implementations focus primarily on chatbot-style question answering, providing text-based responses without deeper use of on-site spatial context, or learning progression [3-5]. Consequently, both approaches fall short in enabling integrated, intent-aligned interpretations of cultural heritage within real-world environments.

In this study, we introduce a LLM-enabled mobile AR system that reflects the on-site context, answers user questions in real time, and presents cultural heritage information through quizzes. This system integrates map-based navigation with AR interaction and leverages a structured cultural heritage knowledge base combined

with an LLM to support real-time user queries, generate quizzes, and store user-LLM question-answer interactions, enabling future visitors to browse or learn from the questions asked by others. This approach promotes active learning, collective knowledge building, and situated exploration, allowing users to proactively explore, question, and learn within the actual historic setting.

2 RELATED WORKS

Prior research on LLM-XR integration identifies education as the most active domain, where LLMs enhance learning through real-time analysis of learner input and personalized feedback across XR-based learning environments [2-5]. In cultural heritage, existing work mainly emphasizes personalized interpretation, narrative explanation, and recommendation, as demonstrated by systems such as MAGICAL[3] and CulturAI [4]. More recent studies have explored extending LLM functionality toward environmental manipulation through natural language commands; however, most LLM-XR research remains centered on context understanding and response generation, with limited attention to integrated, collaborative learning frameworks that combine location-based AR exploration, RAG-based QA, quiz-based assessment, and shared knowledge building in outdoor cultural heritage settings.

3 SYSTEM DESIGN

Figure 1 shows the proposed system architecture, which integrates the Cultural Heritage Knowledge Base (CHKB), the LLM-based QA/Quiz Generation Engine, the Backend Server, and the LBS-based Mobile AR Client. These components are linked via event-based communication to deliver an immersive environment that merges spatial awareness, multimodal interaction, and advanced AI reasoning. For a more natural user experience, the system utilizes TTS (Text-to-Speech) and SR (Speech Recognition) technologies to support voice-centric guidance, enhancing accessibility and engagement beyond simple text display. To facilitate context-based commentary rather than mere database lookups, the system is founded upon a structured knowledge base structure.

The Cultural Heritage Knowledge Base (CHKB) serves as the core foundation for QA and quiz. Cultural heritage data from public portals are systematically reorganized into an ontology-driven structure to refine semantic relationships among data elements. This results in each cultural heritage point of interest (POI) being represented as a multi-layered JSON entity that captures spatio-

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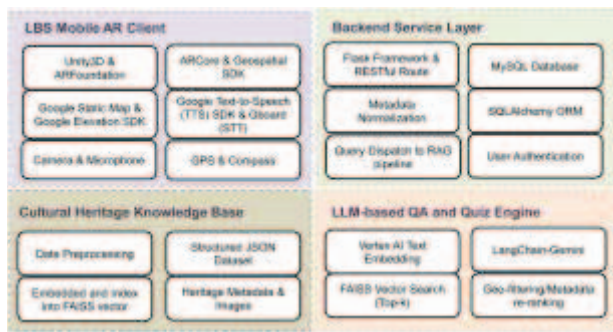


Figure 1: System architecture of LLM-based QA and Quiz Mobile AR for LBS-based cultural heritage exploration.

temporal information, physical attributes, historical context, and symbolic meaning. This structured knowledge base functions as the retrieval source in a retrieval-augmented generation (RAG) pipeline, providing essential factual grounding that enables the LLM to generate reliable, context-specific natural language explanations and robustly support interactive, in-field AR exploration.

The Question Answering and Quiz Generation is designed to provide factual, educational content, leveraging Google Vertex AI, LangChain, and the Gemini LLM. It employs a Retrieval-Augmented Generation (RAG) pipeline where a user's natural language query is vector-embedded, used to retrieve the top-5 relevant documents from the CHKB, and then packaged into a single context block. The LLM is explicitly constrained to generate its response only from this retrieved context, a measure critical for minimizing factual errors (hallucinations). Furthermore, the quiz module automatically generates four-choice multiple-choice questions, complete with questions, options, answers, and explanations in a consistent JSON format, acting as a core engine for transforming heritage data into educational AR content.

The Backend Server acts as the centralized hub, managing the entire system via a Flask-based REST API for real-time communication with the mobile client. It administers the full RAG pipeline (embedding, search, and LLM generation) and is designed to log user questions, quizzes, and temporal data for future analysis. Finally, the Mobile AR Client is implemented using Unity AR Foundation and ARCore Geospatial SDK, integrating GPS, VPS, compass, and other technologies to realize in-situ exploration. A coordinated manager module seamlessly integrates location and orientation tracking with LLM-based knowledge services, seamlessly connecting dynamic multimedia outputs such as text, images, 3D models, and audio narration with the physical world to provide a consistent, field-centric learning experience that enhances user understanding and engagement.

4 INTERACTION SCENARIOS AND EXPERT VALIDATION

As shown in Figure2, the Jongno-gu AR guide integrates spatial proximity, AR overlays, and LLM-driven interpretation to support situated cultural heritage learning. When users approach POI, a POI detail view is automatically displayed, providing contextual information and audio narration, from which users can engage in quiz interactions or natural language question answering. AR quizzes appear as overlays on the live camera view, while spoken or typed questions are processed through a RAG-powered QA pipeline that delivers fact-grounded responses. The system also supports collaborative exploration, allowing multiple users to solve quizzes and share QA responses on-site, thereby fostering collective inquiry. Expert evaluation involving twelve specialists in HCI, XR, software, and cultural heritage domains reported



Figure 2: POI Detail view, QA (two-users), and Quiz (solo-user)

consistently high ratings across semantic accuracy, contextual appropriateness, spatial awareness, explanation quality, and comparative effectiveness, confirming the system's strong factual reliability, contextual relevance, and applicability as a real-world, location-aware cultural heritage guide.

5 CONCLUSION

This study presented an intelligent mobile AR platform integrating RAG-powered LLMs with a structured cultural heritage knowledge base for real-time, location-aware exploration. By combining map navigation, AR-based spatial interaction, and LLM-driven QA, the system transforms passive viewing into active, participatory learning for solo and multi-user scenarios. While field deployments proved feasible, future work will optimize system performance via semantic caching and lightweight, task-specialized LLMs. To expand knowledge coverage reliably, a tiered approach is planned, prioritizing the ontology-based CHKB alongside verified external repositories and expert-in-the-loop validation. Furthermore, controlled field studies with tourists and students will evaluate usability and learning effectiveness to inform design guidelines for deploying AI-augmented XR platforms in real-world heritage tourism.

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